System Development Life Cycle (SDLC)

Detailed Overview, Types, and the Waterfall Model

By: Mehnaz Khan

Learning Objectives

1. Getting familiar with the overall process of developing a new software application/ information systems development.
2. Getting familiar with the System Development Life Cycle (SDLC) phases and types.

Information systems development

- The idea of designing a new business process which may need new hardware or software to support it!
- The idea of adding a new functionality to an existing software system.

- **Information Systems Development**
- **Software Engineering,**
- **Systems Development Life Cycle (SDLC)**
- Those terms are not perfect synonyms, but close cousins.
- The emphasis might be different:
- -SW Engineering with a more technical and practical emphasis.
- -Systems development with more business process emphasis.

Software Engineering != Programming

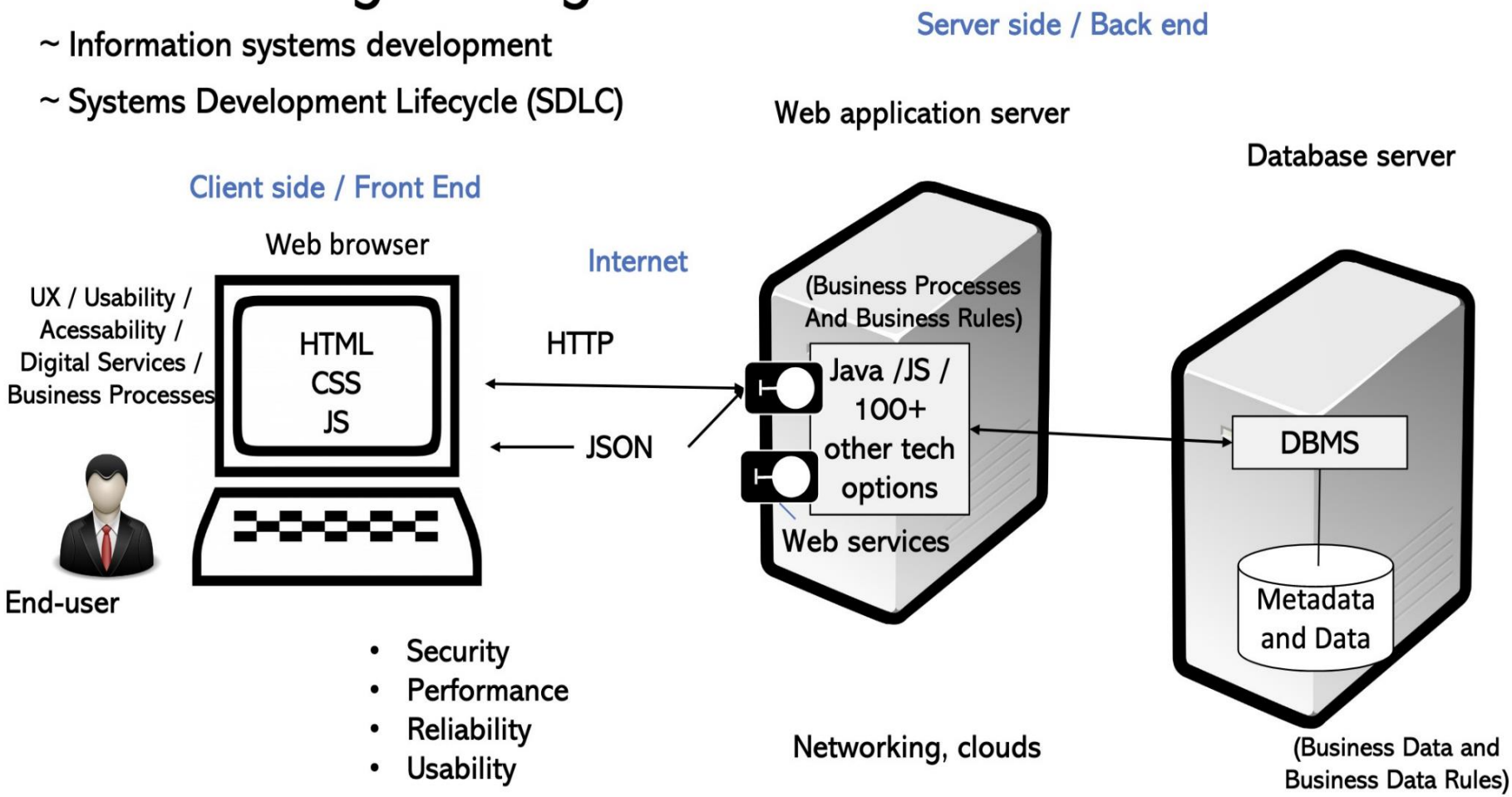
- Programming => Coding
- **A software application** => “a set of instructions or programs that tell a computer what to do”
- **Software engineering** => “a process of analysing user requirements and then designing, building, and testing software application which will satisfy those requirements.”

High-level overview of how a modern web-based information system works

Software engineering

~ Information systems development

~ Systems Development Lifecycle (SDLC)



- Security
- Performance
- Reliability
- Usability
- . . .

©2018 Juhani Välimäki

What is SDLC?

- SDLC stands for Software/System Development Life Cycle.
- **SDLC** is a **general framework** that describes the **entire process of developing software**, from idea to deployment and maintenance.

What is SDLC?

- A structured process to determine how an Information System (IS) supports business needs.
- Involves designing, building, and delivering the system to users

- Sounds simple in theory—complex in real-world execution
- a study conducted in 2008 found success is “improbable” in 68% of technology projects.
- a 2023 report by the Boston Consulting Group found that more than 30% of their organization's technology development projects were over budget and late.
- Some projects are totally abandoned
- Some are delivered to the users significantly late, cost far more than expected, and have fewer features than originally planned.

Why?

- Project failure is rarely due to just one reason—it’s often a combination. E.g
- Unclear or changing requirements
- Poor planning
- Weak communication between developers and stakeholders
- Lack of user involvement and feedback
- Technical issues: bugs, integration problems, outdated tools
- Weak project management

System Development Life Cycle (SDLC)

The SDLC consists of **four main phases**:

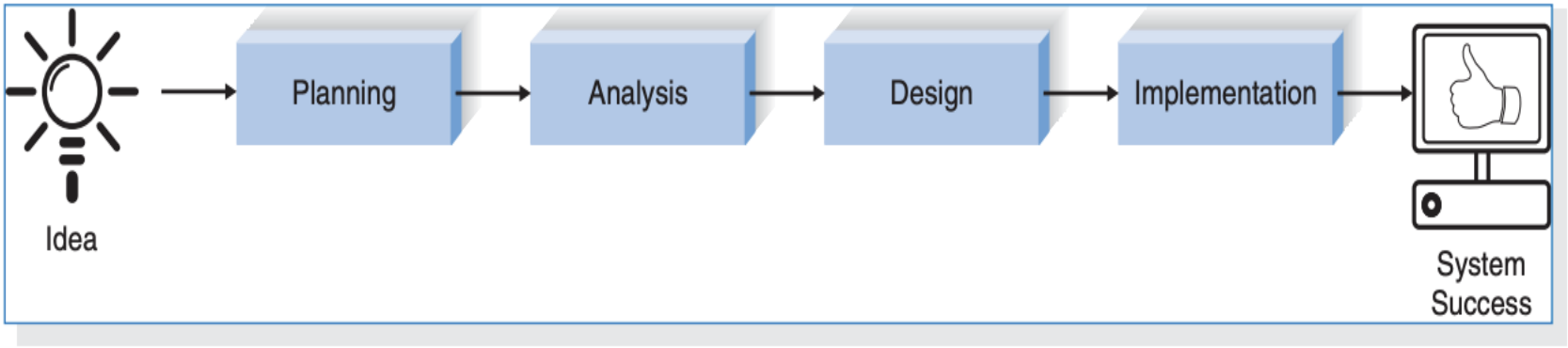


Figure shows SDLC as a **logical, linear sequence**
In **some projects**, phases do follow this straight path, but its not always linear
In **many projects**, phases are applied:
- **Consecutively:** for example in Waterfall model
- **Incrementally**: Incremental model
- **Iteratively:** Iterative model

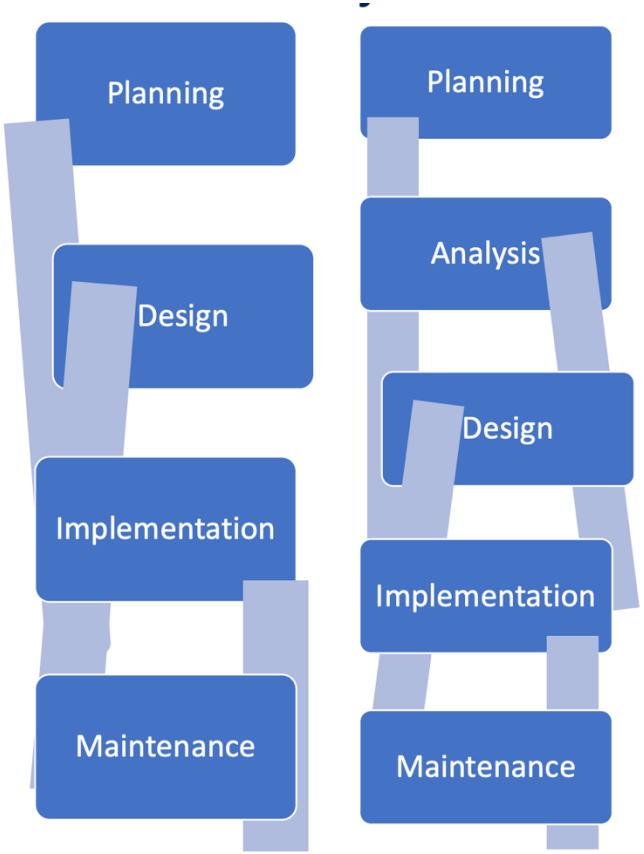
Consecutive Phases are completed one after another with no overlap

Used in: Projects with **fixed, well-defined requirements**

Incremental: The system is built and delivered in small functional parts (increments), adding new features step-by-step.
Used in: Projects needing **early delivery of basic features**, with gradual enhancements, e.g. Educational platforms (start with basic courses, then add assessments, forums)

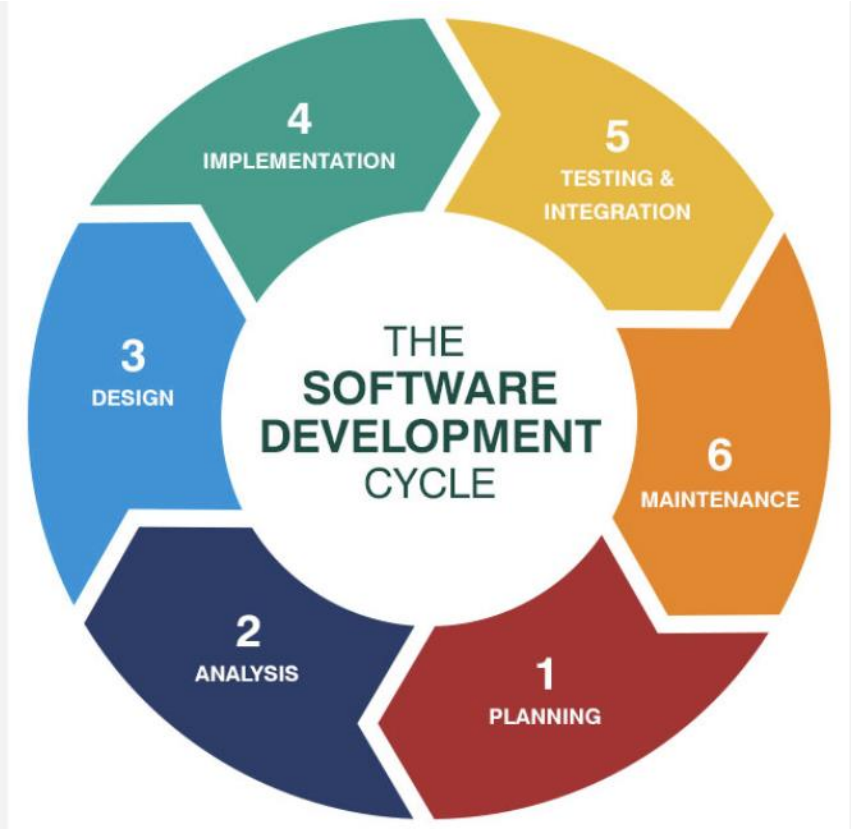
Iterative: System is built in **repeated cycles**—design, build, test, improve through multiple iterations—until it's complete. Build a version, get feedback, refine, and repeat
Used in: Projects where **requirements evolve,** e.g, game development

- SDLC stages are better viewed as **themes or task areas**, not always linear phases
- **Overlap is common** (e.g., Analysis may continue during Design and Implementation)
- SDLC outlines the **processes, methods, tools, and approaches** used within each theme
- The **Implementation phase** can also include **Testing and Deployment** activities



Phases of SDLC

1. Planning/ Feasibility Study
2. Requirements Gathering and Analysis
3. System Design
4. Implementation (Coding)
5. Testing
6. Deployment
7. Maintenance



Each phase includes a **series of detailed steps**

The output of each step is a **deliverable**:

Examples: documents, diagrams, code files, and reports.

Planning

- It starts when the request for a new system is received.
- **System request** outlines the business need and how the system can fulfil that.
- **Purpose**: Understand *why* the system is needed and *how* it will be developed.
- Planning involves **2 steps**:

Step 1: Project Initiation

- Identify the system's **business value** (cost reduction, revenue growth)

  Includes a **Feasibility Analysis**:
- **Technical feasibility** – Can we build it?
- **Economic feasibility** – Will it provide value?
- **Organizational feasibility** – Will it be adopted?
- The **system request** and **feasibility analysis** are presented to an information systems approval committee, which decides whether the project should be undertaken.

Step 2: Project Management

- Begins after project approval
- Project manager:
  - Creates the **work plan**
  - **Staffs the project**
- The **deliverable** for project management is a **project plan** that describes how the project team will go about developing the system.

Deliverables/Output of Planning Phase

1. Feasibility analysis
2. Project plan/work plan

Analysis

Who will use the system, **what** it will do, and **where/when** it will be used

- Focuses on understanding the **current system** and envisioning improvements.
- Analysis has 3 main steps:

1. Develop an Analysis Strategy

- Study the **current system** (*as-is*) and identify issues
- Propose improvements for the **future system** (*to-be*)

2. Gather Requirements

- Use methods like **interviews, workshops, questionnaires**
- Define a **system concept** with help from users and stakeholders
- Create **business analysis models** (data + processes) - describes how the business will operate if the new system were developed.

3. Create a System Proposal

- Combines analysis, system concept, and models into a document called **System Proposal.**
- Presented to project sponsor/decision-makers
- Acts as the **first step in design** (some call it "analysis and initial design")

Deliverables/Output of Analysis Phase

1. Requirement definition
2. System Proposal

Design

Focuses on **how** the system will operate in terms of:

- Hardware, software, and network infrastructure
- User interface, forms, and reports that will be used
- Specific Programs, databases, and files that will be needed.
- It has 4 steps:

1. Determine the Design Strategy

Decide whether to:

- Develop the system by the company's own programmers.
- Or to outsource to another firm
- Or buy an existing software

2. Develop Architecture & Interface Design

- Architecture Design:

Define hardware, software, and network needs

- Interface Design:

Design user navigation (menus, buttons), forms, and reports that will be used.

3. Develop the Database & File Specifications

- Identify exactly what data will be stored and where they will be stored.

4. Develop the Program Design

- The analyst team defines the programs that need to be written and exactly what each program will do.

Deliverables/Output of Design Phase

- Collection of deliverables (architecture design, interface design, database and file specifications, and program design) is the **System specification.**
- It is used by the development team during implementation.
- Project plan and feasibility are **re-evaluated**
- Approval committee decides whether to **continue or terminate** the project

Implementation

- Final phase where the system is built (or purchased), installed, and supported.
- Receives the **most attention** due to its complexity and impact because its often the **longest and most expensive** phase in SDLC
- It has 3 steps:

1. System Construction

- Build the system and conduct thorough **testing**
- Goal: Ensure system functions as designed
- Testing is **critical** and often takes more effort than coding.

Unit testing- individual parts of the code for errors

System testing- check if the components work together properly

User acceptance testing- Test by real system's users to ensure that it meets their standards

2. System Installation/Deployment

- Replace the old system with the new one
- Several approaches may be used to convert from the old to the new system.
- Includes **data migration** and **user training-**teach users how to use the new system and help manage the changes caused by the new system
- Different **conversion approaches** (parallel, direct cutover)

3. Establish a Support Plan

- Plan for **post-implementation review**
- Define how to identify major and minor **system changes needed.**

Types of SDLC Models

1. Waterfall Model
2. V Model
3. Iterative Model
4. Incremental Model
5. Agile Model- use a **hybrid of both**: developing in **increments** and improving **iteratively.**

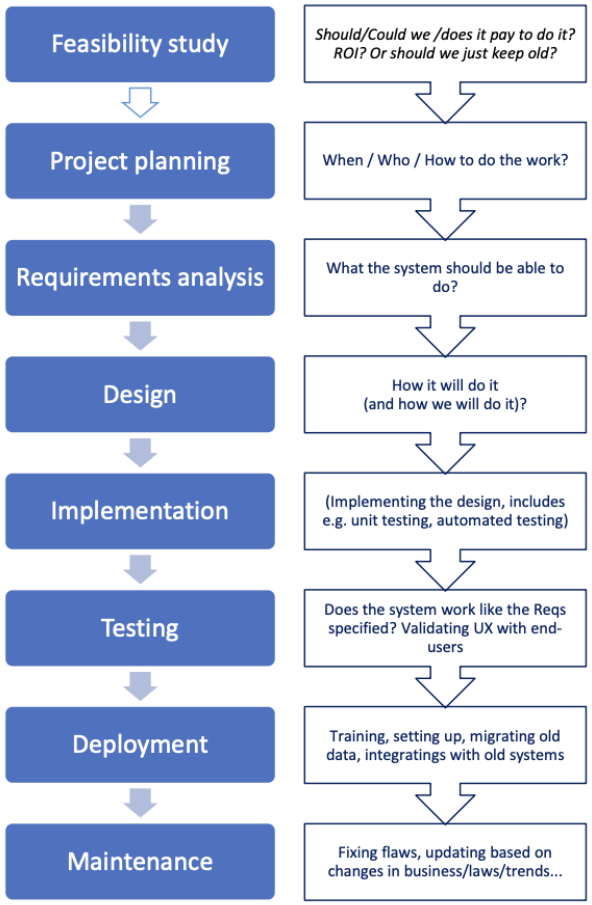
What is the Waterfall Model?

- Oldest and most traditional SDLC model.
- Sequential approach: each phase must be completed before the next.
- Each stage relies on information from the previous stage
- Well-suited for projects with fixed requirements.
- Difficult to go back to a previous phase once completed.

Stages of the Waterfall Model

1. Feasibility Study
2. Requirements Gathering and Analysis
3. System Design
4. Implementation (Coding)
5. Integration and Testing
6. Deployment
7. Maintenance

Each stage flows into the next like a waterfall.



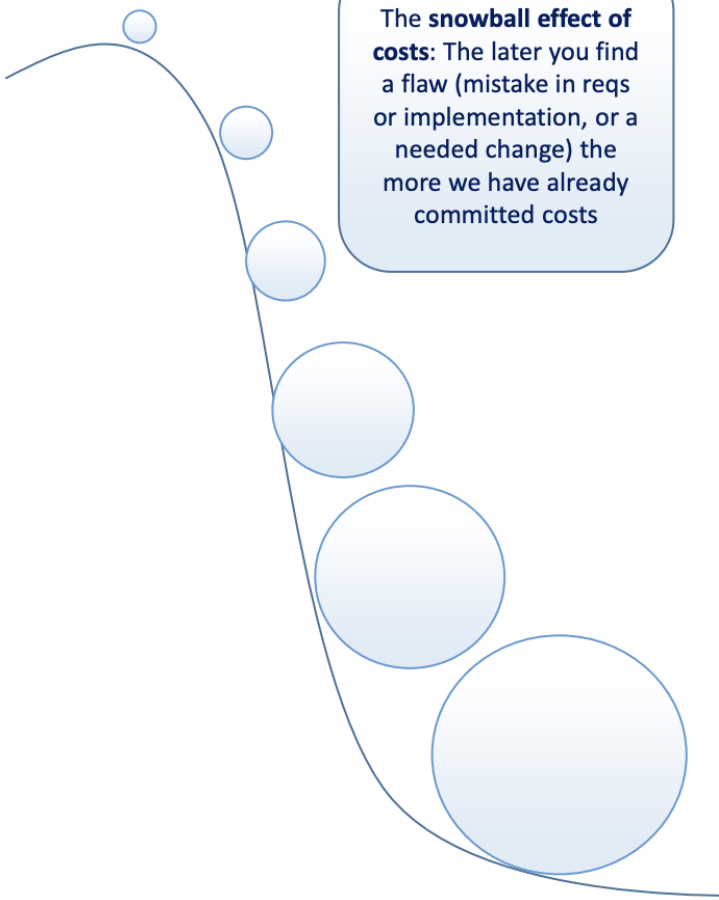
Plan-driven approach (**vs. Agile** development)

Everything is planned before-hand.

E.g. the whole project scope/schedule/budget in Project planning. Or most of the Implementation in Design phase

Roughly 10-50% of the total costs used for the actual implementation.

In many projects even as much as 50% of the total costs used in maintenance!



The **snowball effect of costs**: The later you find a flaw (mistake in reqs or implementation, or a needed change) the more we have already committed costs

©2018 Juhani Välimäki

small issues in early stages can grow into much larger problems as the project progresses

In Waterfall, each phase is done sequentially.

If there's a mistake, misunderstanding, or missing requirement early on, it might not be discovered until much later (usually during **testing**).

At that point, fixing it is expensive and time-consuming, because:

- Design might need to be redone
- Code might need to be rewritten
- Schedules and costs go off track
- This is one reason why **Agile and Iterative models** are preferred in dynamic projects

Advantages of Waterfall Model

- Simple and easy to understand.
- Clear structure and well-defined stages.
- Good for small or well-defined projects.
- Progress is measurable and predictable.

Disadvantages of Waterfall Model

- Inflexible to changing requirements.
- Difficult to go back to a previous stage.
- Not suitable for complex or evolving projects.
- Delayed testing phase increases risk.

Project Planning Constraints

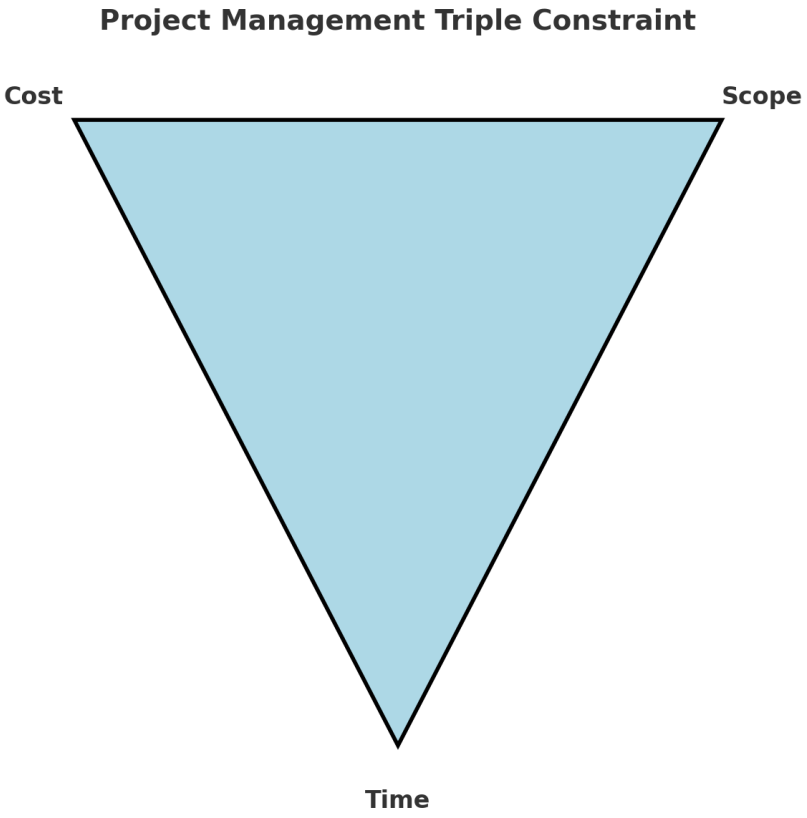
- **3 main project planning constraints**—often called the **"Triple Constraint"**

Time- The project's **schedule** and deadlines (durations, delivery dates)

Cost- The **budget** and resources allocated

Scope-capabilities/features/properties with which the ready system would support the business

- Defines what is *included* and *excluded* from the project



These three are **interdependent**—changing one affects the others:
If you want to expand scope, you'll need more time or cost.

A stable triangle = a well-balanced project

- Distorting one side = risk to quality, delivery, or expectations.
- In plan-driven approach you typically need to fix all of this and hope to get lucky.
- Agile development, on the other hand, has a different idea…

Discussion

- What problems might arise because of the plan-driven and do-all-before-advancing-to-next-phase approach? Think e.g. about the customers, development team, technology and complications, and the market.
- Is the Waterfall development process model used at all today? If it could be used, then in which kind of projects for which kind of customer organizations?